PASS SCIENCE ASSIGNMENT

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Backpropagation Derivation:

1. **Initial values:**

$w = 0.5,\ b = 0,\ v = 1,\ c = 0$

$x = 1,\ t = 2,\ \eta = 0.1$

2. **Forward propagation:**

$h = \tanh(w + b) = \tanh(0.5) = 0.462117$

$\hat{y} = vh + c = 0.462117$

$e = \hat{y} - t = 0.462117 - 2 = -1.537883$

3. **Loss calculation:**

$L = \frac{1}{2}(\hat{y} - t)^2$

$\boxed{L = 1.182542}$

4. **Gradient calculation:**

$\partial L = eh = -0.710682$

$\frac{\partial L}{\partial c} = e = -1.537883$

$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial b} = ev(1 - h^2) = -1.209464$

5. **Weight updates**

using the formula:

$$P_{new} = P_{old} - \eta \frac{\partial L}{\partial p}$$

Parameter	updated Value
w	0.620946
b	0.120946
v	1.071068
c	0.153788

6. Verification:

Analytical gradients and finite difference gradients were almost identical, confirming the correctness of the implementation.

Conclusion:

The program successfully performs forward propagation, calculates gradients using the chain rule, and updates parameters using gradient descent.